Inserting a short chord at any two positions gives Knuth equivalent matchings

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Editorial note. The counted statement is the author's labelled lemma environment at publisher-source.tex lines 1157-1160; its proof environment opens at line 1161, immediately after \end{lemma}, so the pairing is by adjacency and the retained proof range 1161-1165 is the whole environment.

Editorial note. The definition of the insertion (source lines 1141-1151) is retained as context because the counted proof appeals to it at source line 1164; it is the only outside label of the retained proof.

Editorial note. This article carries no label-number dump and the retained counter values are left empty for the whole source, so the printed numbering of the card is the wrapper's default; the identity of the counted item is its author label lem:insert i equivalent insert j.

Editorial note. The retained preamble is the editorially prepared standalone preamble of this source.

Editorial scope. One original result of the pinned publisher article, counted once: Knuth equivalence of the matchings obtained by inserting a short chord at two different positions (author label lem:insert i equivalent insert j).

Allowed standard prerequisites (editorial).

- Insertion of a short chord. insrt_i(m) is obtained by inserting a short chord joining the vertices i and i+1 and shifting all later vertices by two.. It is the operation whose two instances are compared.
- Knuth equivalence of matchings. Two matchings are Knuth equivalent when one is obtained from the other by a sequence of elementary Knuth transformations.. It is the relation the lemma establishes between the two insertions.
- Reduction process and elementary transformation. An elementary Knuth transformation is the local rewriting of three consecutive chords used to generate Knuth equivalence.. The induction step produces such a transformation explicitly.

Definition 0.1. Let $m \in \mathcal{M}_N$ be a matching, and let $1 \leq i \leq N+1$ be an index. Denote by $\mathrm{insert}_i(m) \in \mathcal{M}_{N+2}$ the matching obtained by inserting a short chord that matches the vertices i and $i+1$, while pushing every vertex $j \geq i$ to position $j+2$. Formally, denote by $f_i : [N] \to [N+2] \setminus \{i, i+1\}$ the function that is described as follows:

$$f_i(j) = \begin{cases} j & \text{if } j < i, \\ j+2 & \text{if } j \geq i. \end{cases}$$

Then the matching $\mathrm{insert}_i(m)$ consists of the chords $(f_i(j_1), f_i(j_2))$ for all $(j_1, j_2) \in m$ together with $(i, i+1)$, and consists of the unmatched vertices $(f_i(j))$ for all $(j) \in m$.

Lemma 0.2. Let $m \in \mathcal{M}_N$ be a matching and let $1 \leq i, j \leq N+1$ be indices. Then $\mathrm{insert}_i(m)$ is Knuth equivalent to $\mathrm{insert}_j(m)$.

Proof. We may assume without loss of generality that $i \leq j$, and prove the statement by induction on $j - i$. The statement is obvious for $i = j$.

Assume that $i < j$. By Definition 0.1, the matching $\mathrm{insert}_{i+1}(m)$ is Knuth equivalent to $\mathrm{insert}_i(m)$. By the induction hypothesis, $\mathrm{insert}_{i+1}(m)$ is Knuth equivalent to $\mathrm{insert}_j(m)$ as well. Therefore, $\mathrm{insert}_i(m)$ is Knuth equivalent to $\mathrm{insert}_j(m)$. □